



System Block Diagram — Seed Stage

High-level architecture of the RaraPLC main board: protected/filtered power chain, the STM32H743ZIT6 core, field I/O, expansion, and machine interfaces. This is the seed-stage reference diagram for the public repository (github.com/jorgefelipecurio/rarapl) — block content is grounded in the reviewed KiCad schematics (power chain, DI opto channel, DO low-side channel).

seed / draft grounded in reviewed schematics

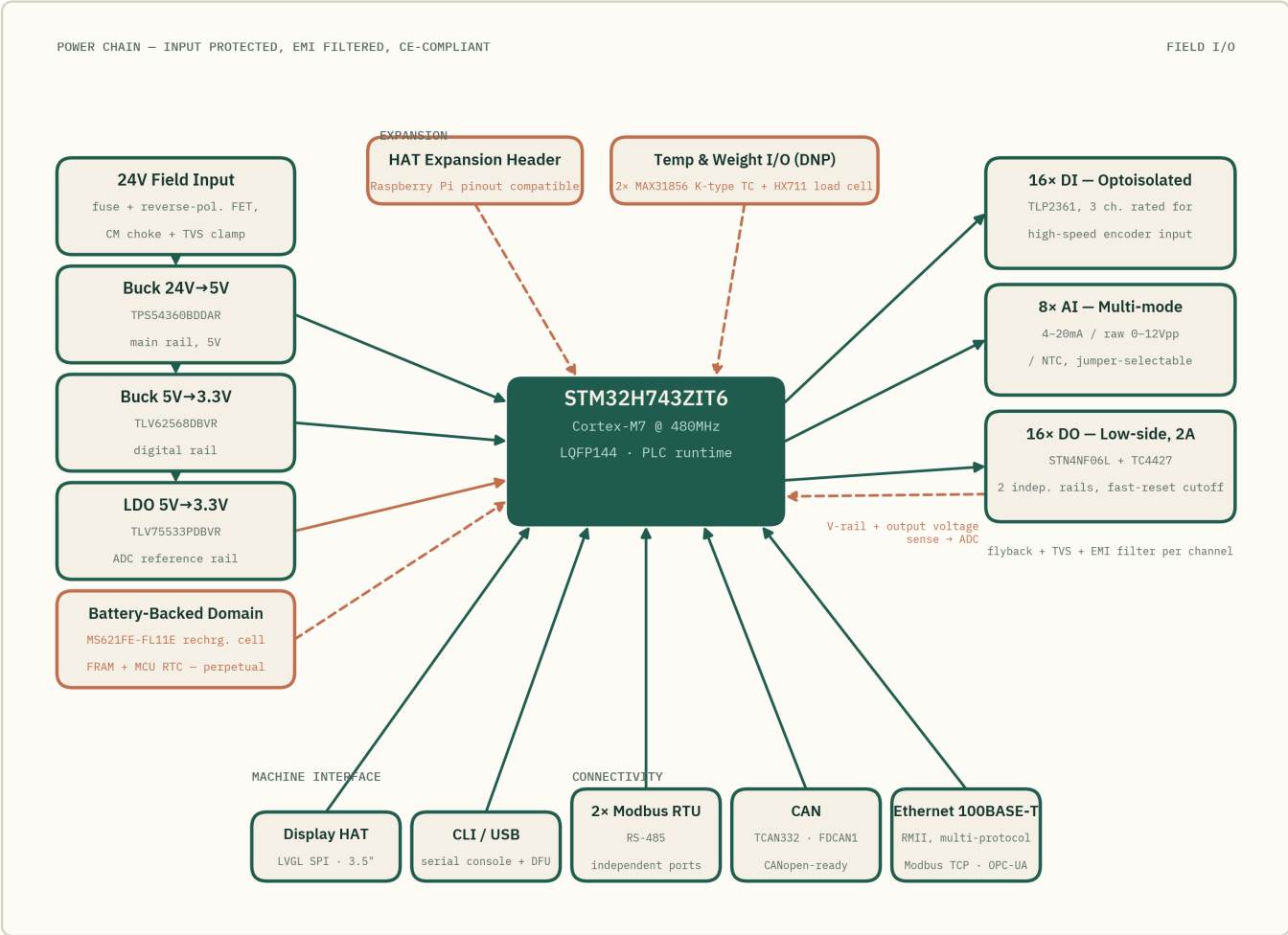


FIG. 1 — System-level block diagram. Green arrows: power/data. Dashed terracotta arrows: expansion buses and diagnostic sense lines.

DESIGN NOTES

- Input stage is reverse-polarity and EMI protected (fuse, P-MOSFET ideal diode, common-mode choke, TVS clamp) for CE-compliant conducted/radiated emissions behavior.
- FRAM and the MCU's RTC sit on a separate battery-backed domain (rechargeable cell), independent from the main 3.3V/5V rails — retention survives a full power-down.
- All 16 DO channels are low-side switched, split across two independent V-rails, with per-channel flyback + TVS protection and hardware-assisted fast-reset cutoff.
- Both DO output voltage and V-rail voltage are read back into the MCU's ADC for diagnostics, independent of the DO drive path.
- The temperature/weight I/O (2× MAX31856 K-type thermocouple + HX711 load-cell input) is a DNP (do-not-populate) footprint on the main board — populated only if the application needs it.
- Connectivity: 2× independent RS-485 Modbus RTU ports, 1× CAN port (TCAN332 on FDCAN1, CANopen-ready), 1× 100BASE-T Ethernet port, multi-protocol (Modbus TCP, OPC-UA).

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